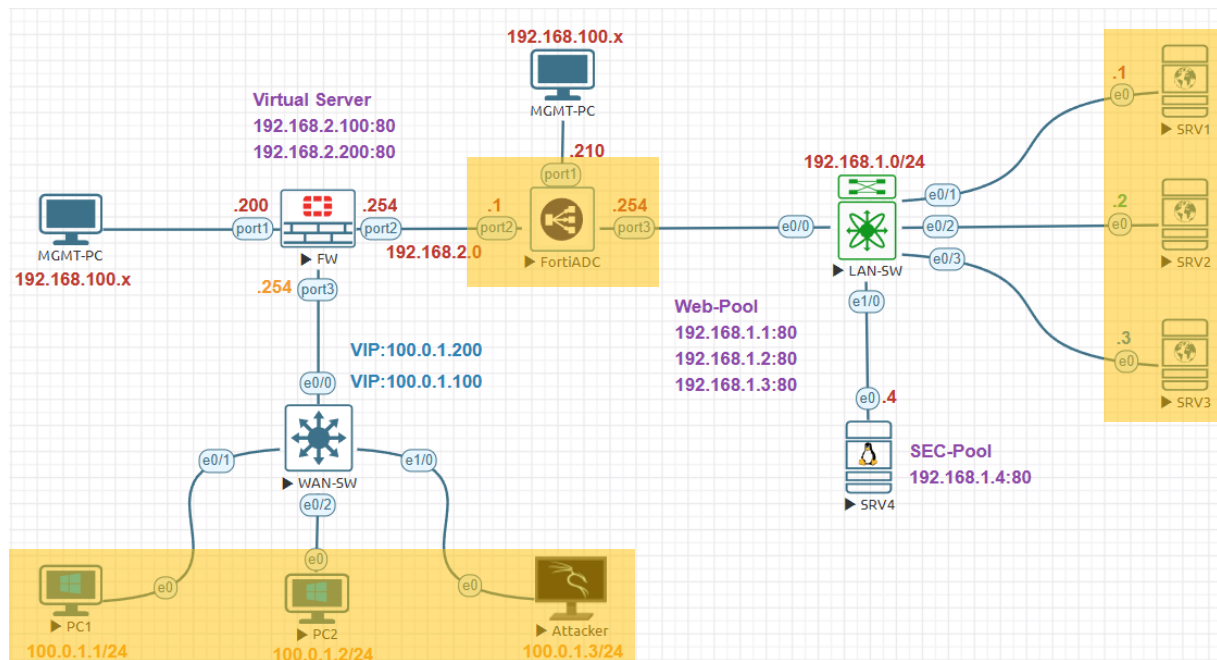


SSL Offload Lab:



Management Subnet	192.168.100.0/24
FortiGate Management IP	192.168.100.200
FortiADC Management IP	192.168.100.210
Internal Servers Subnet	192.168.1.0/24
FortiADC External Subnet	192.168.2.0/24
FortiGate Firewall External	100.0.1.0/24
FortiADC Internal IP Address	192.168.1.254
FortiADC External IP Address	192.168.2.1
FortiGate Firewall Internal IP	192.168.2.254
FortiGate Firewall External IP	100.0.1.254
SRV1 IP Address	192.168.1.1
SRV2 IP Address	192.168.1.2
SRV3 IP Address	192.168.1.3
SRV3 IP Address	192.168.1.4
Virtual Sever-1	192.168.2.100
Virtual Sever-2	192.168.2.200
FortiGate VIP-1	100.0.1.100
FortiGate VIP-2	100.0.1.200
External PC1 IP Address	100.0.1.1
External PC2 IP Address	100.0.1.2
External PC3 IP Address	100.0.1.3

Import Server's Signed Digital Certificate:

Click **System > Manage Certificates > Local Certificate**. Click **Import**, and then enter the following settings:

The screenshot shows the Fortinet web interface. On the left, the 'System' menu is expanded, and 'Manage Certificates' is selected. The main content area shows the 'Local Certificate' configuration page. The 'Local Certificate' tab is selected, and the 'Import' button is highlighted. Below the buttons is a table of existing certificates:

Name	Subject
Factory	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Fc
Fortiadc_ssl	/CN=FADV0000000TRIAL/O=Fortinet Ltd./OU=Forti/
SSLPROXY_LOCAL_CA	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Fc
HTTP2_RSA_2048	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Fc

Showing 1 to 4 of 4 entries 0 rows selected Show 25 entries

Type: Certificate

Certificate Name: Local-Certificate

Input Type: Manual Input

Certificate: Copy and Paste into the field

Key: Copy and Paste into the field

Password: fortinet

The screenshot shows the Fortinet web interface for the 'Local Certificate' configuration page. The 'Type' is set to 'Certificate', 'Certificate Name' is 'Local-Certificate', 'Input Type' is 'Manual Input', 'Certificate' and 'Key' fields contain the certificate and key data, and 'Password' is 'fortinet'.

Type: Certificate

Certificate Name: Local-Certificate

Input Type: Upload Manual Input

Certificate: +/0fzHonQkkxw2K1Jc/RR
bQFcG7EMSQu3cdsUk57vmLU1Cw==
-----END CERTIFICATE-----

Key: 23T9OMSCsV3fNTXB9DWNViNKA9S1/p8DnkPuxJkJy2L
+U5L7resOErl3T08=
-----END RSA PRIVATE KEY-----

Password: fortinet

Save Cancel

Click **Save**.

Finally, Local certificate has been imported into FortiADC.

Local Certificate Group

Local Certificate

Intermediate CA Group

Intermediate CA

OCSP Response

+ Generate

+ Import

+ Add Filter

Name	Subject
Factory	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=FortiADC/CN=FortiADCVM/emailAddress=support@fortin
Fortiadc_ssl	/CN=FADV0000000TRIAL/O=Fortinet Ltd./OU=FortiADC/L=Sunnyvale/ST=California/C=US/emailAddress=sup
SSLPROXY_LOCAL_CA	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=FortiADC/CN=Forward_Proxy_Template_CA/emailAddress=
HTTP2_RSA_2048	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=FortiADC/CN=HTTP2_RSA_2048/emailAddress=support@
Local-Certificate	/C=CA/ST=Ontario/L=Ottawa/O=Fortinet/OU=FortiADC Labs/CN=FortiADC Labs/emailAddress=training@for

Showing 1 to 5 of 5 entries0 rows selectedShow 25 entries

Local Certificate Group:

Click **System > Manage Certificates > Local Certificate Group**. Click **Create New**, and then enter the following setting:

System

Settings

High Availability

Traffic Group

Administrator

SNMP

Replacement Messages

FortiGuard

Debug

WCCP

CERTIFICATE

Manage Certificates

Local Certificate Group

Local Certificate

Intermediate CA Group

Delete

Create New

Add Filter

Name

LOCAL_CERT_GROUP

HTTP2_CERT_GROUP

Showing 1 to 2 of 2 entries 0 rows selected Show 25 entries

Group Name: Local-Certificate-Group

Click **Save**.

Local Certificate Group

Group Name

Group Member

Delete Create New Add Filter

ID	Local Certificate	OCSP Stapling	Intermediate CA Group
No data available in table			

Showing 0 to 0 of 0 entries 0 rows selected Show entries

Save Cancel

In the **Group Member** pane, click **Create New**, and then enter the following settings:

Default: On

Local Certificate: Local-Certificate

Local Certificate Group

Group Name

Group Member

Delete Create New Add Filter

ID	Local Certificate	OCSP Stapling	Intermediate CA Group
No data available in table			

Showing 0 to 0 of 0 entries 0 rows selected Show entries

Save Cancel

Local Certificate Group

Group Member

Default ☒

Local Certificate

OCSP Stapling

Intermediate CA Group

Extra Certificate

Save Cancel

Click **Save**, and then click **Save** again.

Finally, Local Certificate Group has been created also the local certificate has been added.

Local Certificate Group

Group Name

Group Member

Delete

+ Create New

+ Add Filter

ID	Local Certificate	OCSP Stapling	Intermediate CA Group
1	Local-Certificate		

Showing 1 to 1 of 1 entries 0 rows selected Show 25 entries

Save

Cancel

Create HTTPS Profile:

Click **Server Load Balance > Application Resources > Application Profile**. Click on Type HTTPS name LB_PROF_HTTPS profile to clone it.

Dashboard

Security Fabric

FortiView

System

Shared Resources

Network

Server Load Balance

Virtual Server

Application Resources

Application Optimization

Real Server Pool

Scripting

SSL-FP Resources

Link Load Balance

Global Load Balance

Web Application Firewall

Network Security

Application Profile

Client SSL

HTTP2 Profile

HTTP3 Profile

Delete

+ Create New

+ Add Filter

Name	Type	
HTTP-Profile	HTTP	
LB_PROF_DIAMETER	Diameter	
LB_PROF_DNS	DNS	
LB_PROF_EXPLICIT_HTTP	EXPLICIT_HTTP	
LB_PROF_FTP	FTP	
LB_PROF_HTTP	HTTP	
LB_PROF_HTTP2_H2	HTTPS	
LB_PROF_HTTP2_H2C	HTTP	
LB_PROF_HTTP3	HTTPS	
LB_PROF_HTTP_SERVERCLOSE	HTTP	
LB_PROF_HTTPS	HTTPS	

Name: HTTPS_Profile

Dashboard >	Application Profile	
Security Fabric >	Name	HTTPS_Profile
FortiView >	Type	HTTPS
System >	Specifics	
Shared Resources >	Client Timeout	50 Default: 50 Range: 1-86400 seconds
Network >	Server Timeout	50 Default: 50 Range: 1-86400 seconds
Server Load Balance >	Connect Timeout	5 Default: 5 Range: 1-86400 seconds
Virtual Server	Queue Timeout	5 Default: 5 Range: 1-86400 seconds
Application Resources		
Application Optimization		
Real Server Pool		

Keep the default values for the other settings, and then click **Save**.

Create Client SSL Profile:

Click **Server Load Balance > Application Resources > Client SSL**. Click **Create New** to create a new profile using the following settings:

Dashboard >	Application Profile		Client SSL	HTTP2 Profile	HTTP3 Profile
Security Fabric >	Delete Create New Add Filter				
FortiView >	Name		Customized SSL Ciphers		
System >	LB_CLIENT_SSL_PROF_DEFAULT		disable		
Shared Resources >	LB_CLIENT_SSL_PROF_FORWARD_PROXY		enable		
Network >	LB_CLIENT_SSL_PROF_HTTP2		disable		
Server Load Balance >	Showing 1 to 3 of 3 entries 0 rows selected Show 25 entries				
Virtual Server					
Application Resources					

Name: SSL-Profile

Local Certificate Group: Local-Certificate-Group

Keep the default values for the remaining settings, and then click **Save**.

Dashboard >

Security Fabric >

FortiView >

System >

Shared Resources >

Network >

Server Load Balance >

Virtual Server

Application Resources

Application Optimization

Client SSL

Name

SSL-Profile

Customized SSL Ciphers Flag

AES256-SHA

AES128-SHA

RC4-SHA

RC4-MD5

DES-CBC3-SHA

DHE-RSA-AES256-SHA

DHE-RSA-AES128-SHA

EDH-RSA-DES-CBC3-SHA

Forward Proxy

Client SNI Required

Local Certificate Group

Local-Certificate-Group

Reject OCSP Stapling with Missing Nextupdate

Renegotiation

SSL DH Parameter Size

1024 Bits 2048 Bits 4096 Bits

RFC 7919 Comply

Dynamic Record Sizing

Save

Cancel

Finally, new Client SSL Profile has been created and Local-Certificate-Group has been called.

Application Profile	Client SSL	HTTP2 Profile	HTTP3 Profile	LB Method	Persistence	Error
Delete Create New Add Filter						
Name	Customized SSL Ciphers Flag	Client Certificate Verify				
LB_CLIENT_SSL_PROF_DEFAULT	disable					
LB_CLIENT_SSL_PROF_FORWARD_PROXY	enable					
LB_CLIENT_SSL_PROF_HTTP2	disable					
SSL-Profile	disable					

Configure Virtual Server to Use SSL Certificate:

Click **Server Load Balance > Virtual Server > Virtual Server**. Click on **clone** to clone **WEB-VS**.

Name	Type	Address	Port	Profile	Status	Availability	Settings
WEB-VS	Layer 7	192.168.2.100	80	LB_PROF_HTTP	Enable	✓	
SEC-VS	Layer 7	192.168.2.200	80	LB_PROF_HTTP	Enable	✓	

Name: HTTPS-VS

Disable both **Content Routing** and **Content Rewriting** if enable

Virtual Server

Basic | General | Security | Application Optimization | Monitoring

Name: **HTTPS-VS**

Type: **Layer 7** | Layer 4 | Layer 2

Status: **Enable** | Disable | Maintain

Address Type: **IPv4** | IPv6

Traffic Group: default

Comments: Specify the comments

Specifics

Schedule Pool: ☐

Content Routing: ☐

Content Rewriting: ☐

On the **General** tab, configure the following settings:

Address: 192.168.2.100

Port: 443

Interface: Port2

Profile: HTTPS_Profile

Client SSL Profile: SSL-Profile

Method: LB_METHOD_ROUND_ROBIN

Real Server Pool: WEB-Pool

Click **Save**.

Virtual Server

Basic

General

Security

SSL Traffic Mirror

Application Optimization

Monitoring

Configuration

Address

192.168.2.100

Example: 192.0.2.1

Port

443

Default: 80 Range: 0 or 1-65535. You can specify up to eight ports or port ranges separated by space, e.g., 80-90 100. Valid values are from 0 to 65535, with 0 for Layer-4 virtual servers only.

Connection Limit

0

Default: 0 Range: 0-100000000 concurrent connections

Interface

port2

Resources

Profile

HTTPS_Profile

Client SSL Profile

SSL-Profile

Persistence

Click to select.

Method

LB_METHOD_ROUND_ROBIN

Real Server Pool

WEB-Pool

Clone Pool

Click to select

Auth Policy

Click to select

Finally, another Virtual Server has been clone from WEB-VS however the port and profile are different.

Virtual Server

Content Rewriting

Content Routing

NAT Source Pool

Schedule Pool

Delete

+ Create New

Search

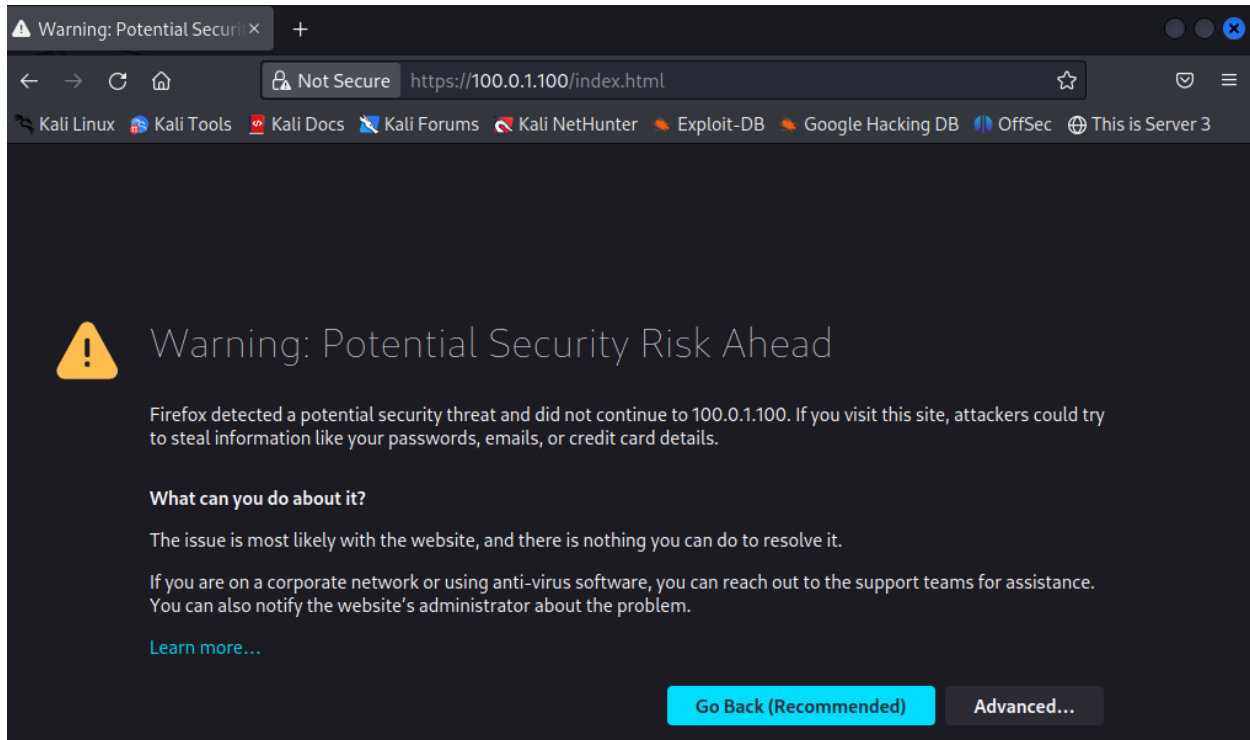
+ Add Filter

Name	Type	Address	Port	Profile	Status	Availability	
WEB-VS	Layer 7	192.168.2.100	80	LB_PROF_HTTP	Enable	✓	  
SEC-VS	Layer 7	192.168.2.200	80	LB_PROF_HTTP	Enable	✓	  
HTTPS-VS	Layer 7	192.168.2.100	443	HTTPS_Profile	Enable	✓	  

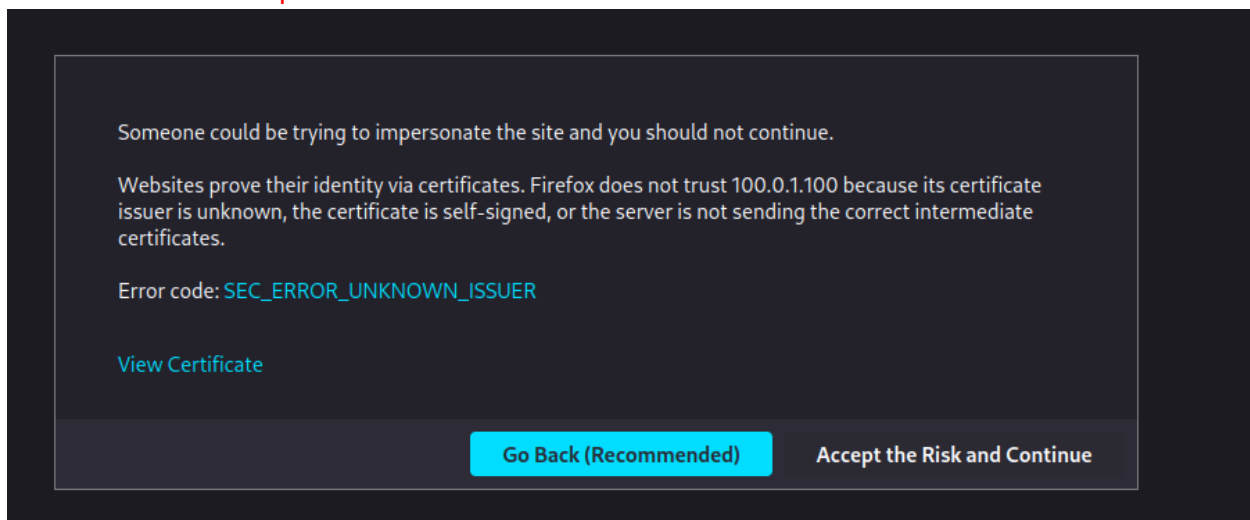
Test HTTPS Connection:

On the External Kali Linux, open a new browser tab, and then connect to the virtual server at <https://100.0.1.100/index.html> .

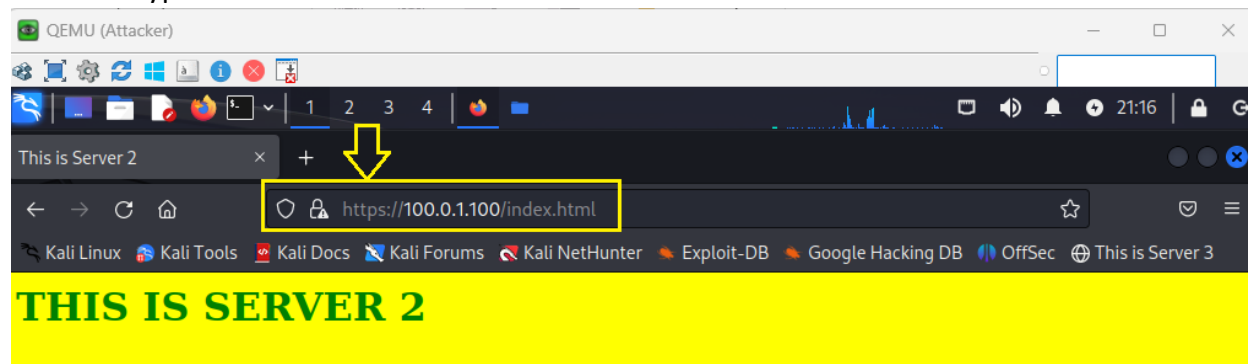
Accept the security certificate warning if it appears, click **Advanced**.



And then select **Accept the Risk and Continue**.



Press **Ctrl+Shift+R** to refresh the browser page several times. Observe that the responses are from each server in the server pool. The communication between your browser and the FortiADC is now encrypted—the communication between FortiADC and the application servers is not encrypted.



FortiADC # diagnose sniffer packet any "tcp and port 443 or port 80" 4

```
FortiADC # diagnose sniffer packet any "tcp and port 443 or port 80" 4
interfaces=[any]
filters=[tcp and port 443 or port 80]
6.934596 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: syn 611073652
6.934906 port2 out 192.168.2.100.443 -> 100.0.1.3.56840: syn 4007139023 ack
6.936745 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: ack 4007139024
6.943939 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: psh 611073653 ack 4
6.945863 port2 out 192.168.2.100.443 -> 100.0.1.3.56840: psh 4007139024 ack
6.947705 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: ack 4007139165
6.949880 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: psh 611074172 ack 4
6.958018 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: psh 611074223 ack 4
6.958080 port2 out 192.168.2.100.443 -> 100.0.1.3.56840: ack 611074769
6.959320 port3 out 192.168.1.254.22652 -> 192.168.1.3.80: syn 2084671881
6.960716 port3 in 192.168.1.3.80 -> 192.168.1.254.22652: syn 75556347 ack 2
6.960757 port3 out 192.168.1.254.22652 -> 192.168.1.3.80: psh 2084671882 ac
6.962150 port3 in 192.168.1.3.80 -> 192.168.1.254.22652: ack 2084672375
6.966500 port3 in 192.168.1.3.80 -> 192.168.1.254.22652: psh 75556348 ack 2
6.969104 port2 out 192.168.2.100.443 -> 100.0.1.3.56840: psh 4007139165 ack
7.007866 port3 out 192.168.1.254.22652 -> 192.168.1.3.80: ack 75556769
7.011071 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: ack 4007139615
7.082607 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: psh 611074769 ack 4
7.082783 port3 out 192.168.1.254.22652 -> 192.168.1.3.80: psh 2084672375 ac
7.084207 port3 in 192.168.1.3.80 -> 192.168.1.254.22652: psh 75556769 ack 2
7.084230 port3 out 192.168.1.254.22652 -> 192.168.1.3.80: ack 75557216
7.084655 port2 out 192.168.2.100.443 -> 100.0.1.3.56840: psh 4007139615 ack
7.086528 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: ack 4007140091
8.203868 port2 in 100.0.1.3.56840 -> 192.168.2.100.443: psh 611075250 ack 4
```

The first part of the sniffer trace is the HTTPS traffic passing from the client to the FortiADC-Primary **port2**. The **port3** traffic shows the decrypted HTTP traffic passing between the FortiADC and the web server.